

Little Red Dots: Stellar Eruptions in Disguise?

Super-Eddington winds mimic black hole signatures; masses may be intermediate.

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Background

Little Red Dots are compact, red sources from JWST, often interpreted as active galactic nuclei. Their spectra show broad Balmer lines and blackbody-like continua, but their true nature is debated.

Their inferred black hole masses often exceed expectations, suggesting alternative explanations beyond virial motions.

What this paper is selling

This paper draws parallels with Type II_n supernovae and Eta Carinae's Great Eruption, proposing that dense outflows create a pseudo-photosphere. This wind scenario naturally explains LRD spectra without invoking massive black holes.

Key results to remember

- Pseudo-photosphere from dense wind mimics blackbody continuum.
- Broad lines from electron scattering, not virial black hole motion.
- Engine mass constrained to <1e5 solar masses, super-Eddington.

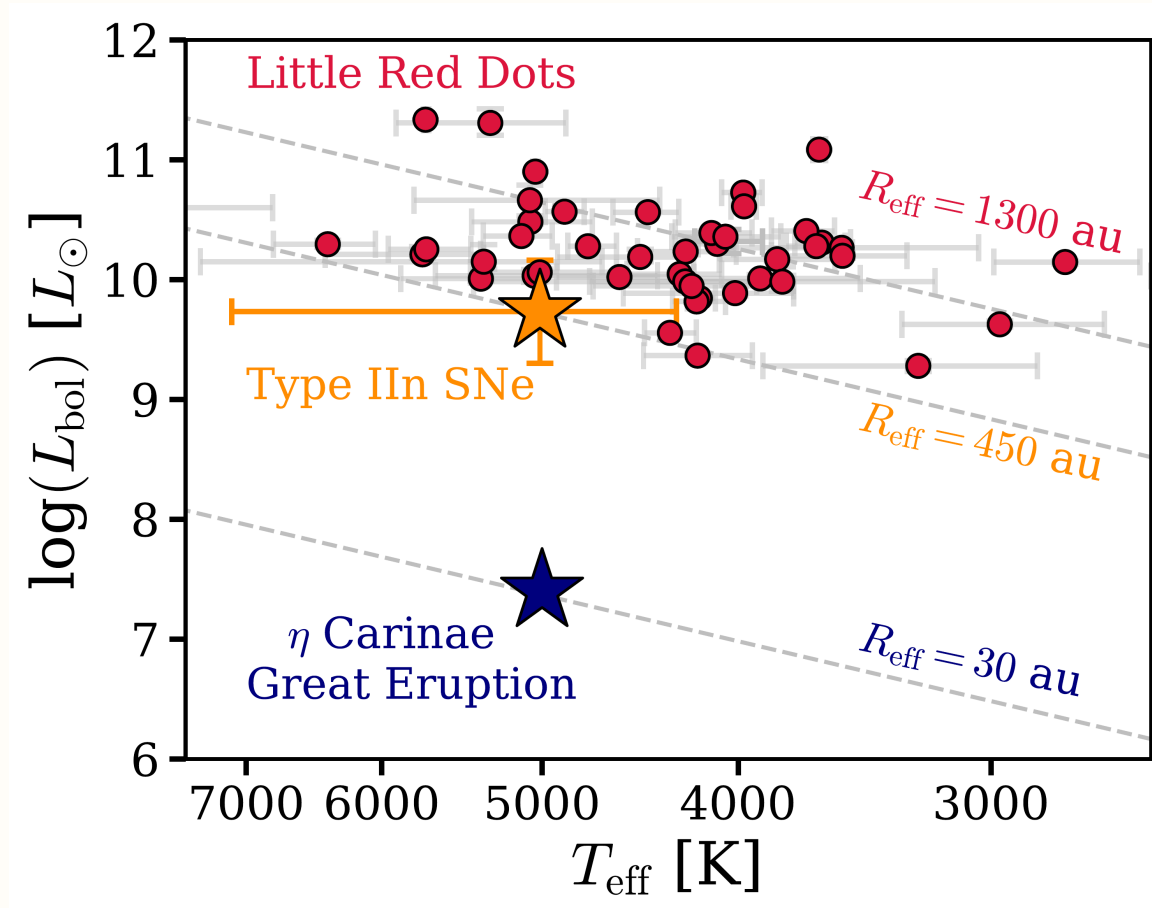


Fig. 4. LRDs, II_n SNe, and Eta Carinae cluster in temperature, suggesting similar pseudo-photospheres.

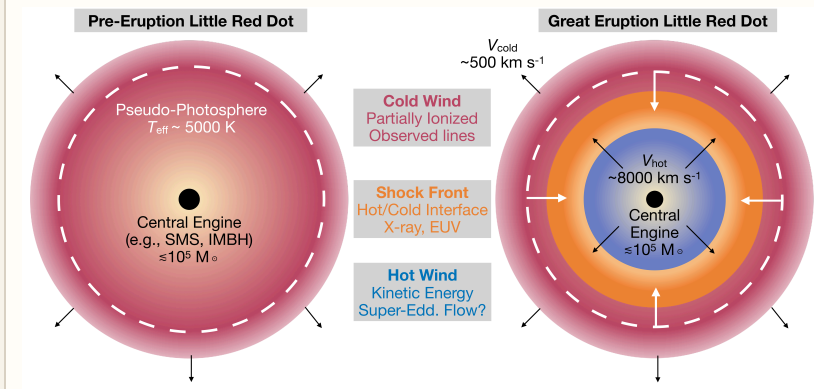


Fig. 8. Schematic: cold wind forms pseudo-photosphere; hot wind shocks create emission lines.